

CHOOSING THE BEST HIGH-END SMARTPHONE

Paromik Chakraborty

Purchasing a premium smartphone usually comes with a number of expectations—superfast performance, butter-smooth interface, high-quality camera and other cool features. Before swiping the debit card, it is a good idea to do a little homework about what makes these phones worth their price. This article provides you with such information and informs you about some of the best choices available in the market today.

What makes these smartphones so powerful

Top-end smartphones are as strong as computers. These carry out a vast range of operations while on-the-move. This is delivered by the mobile phone's processor. Top-end smartphones are expected to come with the latest and most upgraded processor chips. Qualcomm's Snapdragon series is a popular choice, and the flagship chip is Snapdragon 845, which was unveiled towards the end of 2017. Powered by Adreno 630 graphics processing unit (GPU), Kyro 385 central processing unit (CPU) and a powerful modem, the octa-core processor can deliver supercomputing and



data streaming speed. It has a clock-speed of 2.8GHz. The previous version, Snapdragon 835, is also quite powerful. Snapdragon 800 series efficiently deals with overheating on excess usage.

Exynos chips in top-end Samsung devices are quite impressive. Exynos 9810 is the flagship octa-core of the series, with Mali G72 GPU. Four cores clock up to 2.9GHz, while the other four run at 1.9GHz, allowing it to go toe-to-toe with Snapdragon 845. It is interesting to note that both systems have been developed on Samsung Foundry's chip platform. Benchmark tests of various parameters, which reveal the peak performance expected from a chip, have shown a close competition between the two. Exynos shows better performance in terms of CPU and core capacity. However, in terms of graphics, operating system (OS) and Web, Snapdragon barely beats the race. Exynos 8895 chip (9810's predecessor) is also quite powerful.

Huawei launched its flagship processor chip, Kirin 970. The octa-core processor has an artificial intelligence (AI) suite of its own and can clock up to 2.4GHz processing speed. It can be considered on par with Snapdragon and Exynos flagships.

Apple's proprietary flagship chip, A11 bionic processor, beats even Snapdragon 845 in processing capacities. Benchmark scores show that even though Snapdragon has higher caching and memory retrieval speed, A11 is faster in terms of overall practical usage. Additionally, A11 comes with well-developed neural network and machine learning programs. This makes it the processor to beat.

Apart from the processors, higher RAM is also beneficial. The same model of a phone may vary based on RAM specifications. You may find devices that offer 4GB or 6GB of RAM, while some even go up to 8GB. In addition, get one that has higher battery rating (usually in mAh).